CO2-AT1-Part2

Application-Based Questions – TD(0), SARSA & Q-Learning

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Step 1: Define Learning Models and Environment

Let

$$M_1 = \text{TD(0) Learning}$$
$$M_2 = \text{SARSA Learning}$$
$$M_3 = \text{Q-Learning}$$
$$S = \{s_1, s_2, \ldots, s_n\}$$
$$A = \{a_1, a_2, \ldots, a_m\}$$

where

- M_1 = TD(0) Model
- M_2 = SARSA Model
- M_3 = Q-Learning Model
- S = Set of States
- A = Set of Actions

Step 2: Input Data

For each state s,

- Reward:$R(s)$
- Current State:S_t
- Next State:S_{t+1}
- Learning Rate:α
- Discount Factor:γ

- Current Action:A_t
- Next Action:A_{t+1}
- Current Q-value:$Q(S_t, A_t)$

Step 3: Temporal Difference Learning (TD(0))

TD Target

$$TD_{Target} = R + \gamma V(S_{t+1}) \tag{1}$$

Eq. (1) computes the Temporal Difference target using the immediate reward and discounted next-state value.

TD Error

$$TD_{Error} = TD_{Target} - V(S_t) \tag{2}$$

Eq. (2) calculates the prediction error between the target and the current value.

Value Update

$$V(S_t) = V(S_t) + \alpha(TD_{Error}) \tag{3}$$

Eq. (3) updates the value function using the TD error.

Step 4: SARSA Learning

SARSA Target

$$Target = R + \gamma Q(S_{t+1}, A_{t+1}) \tag{4}$$

Eq. (4) computes the SARSA target using the next action selected by the current policy.

SARSA Update

$$Q(S_t, A_t) = Q(S_t, A_t) + \alpha[Target - Q(S_t, A_t)] \tag{5}$$

Eq. (5) updates the action-value function following the on-policy learning strategy.

Step 5: Q-Learning

Q-Learning Target

$$Target = R + \gamma \max_{a} Q(S_{t+1}, a) \tag{6}$$

Eq. (6) calculates the target using the maximum future Q-value.

Q-value Update

$$Q(S_t, A_t) = Q(S_t, A_t) + \alpha[Target - Q(S_t, A_t)] \tag{7}$$

Eq. (7) updates the Q-value using the optimal future action.

Step 6: Performance Evaluation

Learning Accuracy

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} \times 100 \tag{8}$$

Eq. (8) computes the percentage of correct predictions.

Average Reward

$$Average\ Reward = \frac{\sum_{i=1}^{N} R_i}{N} \tag{9}$$

Eq. (9) calculates the average reward obtained during learning.

Mean Squared Error

$$MSE = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{10}$$

Eq. (10) measures the prediction error.

Policy Success Rate

$$Success = \frac{Successful\ Episodes}{Total\ Episodes} \times 100 \tag{11}$$

Eq. (11) computes the percentage of successful episodes.

Convergence Speed

$$CS = N_{episodes} \tag{12}$$

Eq. (12) represents the number of episodes required for convergence.

Training Time

$$TT = t_{end} - t_{start} \tag{13}$$

Eq. (13) calculates the total execution time required for training.

Step 7: Comparative Performance Index

Performance Index

$$PI = w_1 A + w_2 R + w_3 S + w_4 C + w_5 M + w_6 T \tag{14}$$

where:

- A = Learning Accuracy
- R = Average Reward
- S = Success Rate
- C = Convergence Speed
- M = Mean Squared Error
- T = Training Time

Eq. (14) defines the overall performance index using six evaluation parameters.

Weight Assignment

$$w_1 = 0.25,\ w_2 = 0.20,\ w_3 = 0.20,\ w_4 = 0.15,\ w_5 = 0.10,\ w_6 = 0.10 \tag{15}$$

Eq. (15) shows the assigned weights for evaluating TD(0), SARSA, and Q-Learning.

Step 8: Output

- TD(0) Updated State Values
- SARSA Updated Q-values
- Q-Learning Updated Q-values
- Learning Accuracy
- Average Reward

- Policy Success Rate
- Convergence Speed
- Training Time
- Mean Squared Error
- Comparative Performance Index
- Final comparison of **TD(0)**, **SARSA**, and **Q-Learning** using the six evaluation parameters.